

IWIAS

Indian Weather Intelligence and Analytics System

Project Overview

IWIAS is a comprehensive weather monitoring and analytics system designed to collect, store, analyse and visualize weather information. The system combines live weather tracking, historical data management, Indian city comparison, statistical analysis, forecasting, alerts and reporting in a single easy-to-use graphical interface.

Key Features

- Live Weather Tracking
- MySQL Weather Database for storing historical weather observations
- Multi-City Weather Comparison across major Indian cities
- Correlation Analysis for Indian cities to identify relationships between weather parameters • Weather Trend and Comparison Graphs
- Machine-Learning-Based Weather Forecasting (if feasible)
- Weather Alert System for selected conditions and thresholds
- Weather Map Visualization
- Automated Weekly Weather Reports
- PDF Report Export
- Simple and Easy-to-Use Graphical User Interface (GUI)

Project Scope

Multi-city comparison and correlation analysis will focus specifically on major cities in India. Forecasting will be evaluated using historical data and measured against subsequently observed weather conditions where sufficient data is available.

Objective

The objective of IWIAS is to transform raw weather data into useful weather intelligence through database management, visualization, statistical analysis, forecasting and automated reporting.

